

ProPresenter 7 Bilingual Lyrics Adjustment Guide

Overview

This guide explains how to modify and structure bilingual lyrics (Indonesian and English) in ProPresenter 7 decoded presentation files.

File Structure

Main Components

- File Format:** Protocol Buffer text format (decoded from `.pro` binary)
- Key Field:** `rtf_data` - Contains RTF (Rich Text Format) encoded lyrics
- Location:** Inside `cues` → `actions` → `slide` → `presentation` → `base_slide` → `elements` → `element` → `text`

Cue Structure

Each `cue` represents a slide/scene and contains:

```
cues {
  uuid { string: "... " }
  name: "... "
  actions {
    slide {
      presentation {
        base_slide {
          elements {
            element {
              # FIRST TEXT BLOCK (Indonesian)
              rtf_data: "{...Indonesian text...}"
            }
            info: 3
          }
          elements {
            element {
              # SECOND TEXT BLOCK (English)
              rtf_data: "{...English text...}"
            }
            info: 3
          }
        }
      }
    }
  }
}
```

RTF Data Structure

RTF Pattern

```
rtf_data: "{\\rtf0\\ansi\\ansicpg1252{\\fonttbl{\\f0\\fnl FontName;}{\\colortbl{\\...}{\\...formatting...}\\n
osupersub ActualTextHere}"
```

Key Components

Component	Purpose	Example
\\rtf0	RTF version	Always \\rtf0

\\ansi	Character set	Always \\ansi
\\fonttbl	Font table	Defines font used
\\colortbl	Color table	Defines text colors
\\fs104 or \\fs74	Font size	104 = larger, 74 = smaller
\\nosupersub	Text marker	Precedes actual text
}	Closing bracket	End of RTF string

Font Sizes

- `\\fs104` - Large font (52pt) - Usually for first line/Indonesian
- `\\fs74` - Medium font (37pt) - Usually for second line/English

Bilingual Layout Pattern

Standard Two-Line Layout

```
Element 1 (Indonesian):
- Font size: fs104 (52pt) or fs104 (52pt)
- Position: y: 869.1
- Height: 170.72368317470944

Element 2 (English):
- Font size: fs74 (37pt) or fs74 (37pt)
- Position: y: 920.3
- Height: 170.72368317470932
```

Text Positioning

```
bounds {
  origin {
    y: 869.1      # Indonesian text (higher on screen)
  }
  size {
    width: 1920
    height: 170.72368317470944
  }
}

bounds {
  origin {
    y: 920.3      # English text (lower on screen)
  }
  size {
    width: 1920
    height: 170.72368317470932
  }
}
```

Modifying Bilingual Lyrics

Step 1: Identify the Text Block

1. Open `decoded_lyrics_copy.txt`
2. Find the `cue` section you want to modify

3. Locate two consecutive `element` blocks within `base_slide`

Step 2: Locate the RTF Data

Look for the pattern:

```
rtf_data: "{\\rtf0\\ansi\\ansicpg1252...\\nosupersub ActualText}"
```

Step 3: Extract the Text

Text appears AFTER `\\nosupersub` and BEFORE `}`

Example:

```
\\nosupersub Firman terangi jalan ku} ← Indonesian text  
\\nosupersub Your Word illuminates my way} ← English text
```

Step 4: Replace Text

To replace existing text:

1. Keep ALL RTF formatting intact
2. Change ONLY the text after `\\nosupersub`
3. Ensure closing `}` remains at the end

Example:

```
OLD: rtf_data: "{\\rtf0\\ansi\\ansicpg1252{...\\nosupersub Firman terangi jalan ku}"  
  
NEW: rtf_data: "{\\rtf0\\ansi\\ansicpg1252{...\\nosupersub Firman cahaya hidupku}"
```

Complete Example

Before Modification

```
cues {  
  uuid { string: "b1da4c32-6fca-48d7-8e35-d9620b1d414e" }  
  actions {  
    slide {  
      presentation {  
        base_slide {  
          elements {  
            element {  
              bounds { origin { y: 869.1 } size { width: 1920 height: 170.72 } }  
              text {  
                rtf_data: "{\\rtf0\\ansi\\ansicpg1252...\\fs104\\...\\nosupersub Firman terangi jalan ku}"  
              }  
            }  
          }  
          info: 3  
        }  
        elements {  
          element {  
            bounds { origin { y: 920.3 } size { width: 1920 height: 170.72 } }  
            text {  
              rtf_data: "{\\rtf0\\ansi\\ansicpg1252...\\fs74\\...\\nosupersub Your Word illuminates my wa  
y}"  
            }  
          }  
          info: 3  
        }  
      }  
    }  
  }  
}
```

```

    }
  }
}

```

After Modification

```

cues {
  uuid { string: "b1da4c32-6fca-48d7-8e35-d9620b1d414e" }
  actions {
    slide {
      presentation {
        base_slide {
          elements {
            element {
              bounds { origin { y: 869.1 } size { width: 1920 height: 170.72 } }
              text {
                rtf_data: "{\\rtf0\\ansi\\ansicpg1252...\\fs104\\...\\nosupersub Cahaya terangi perjalanan}"
              }
            }
            info: 3
          }
          elements {
            element {
              bounds { origin { y: 920.3 } size { width: 1920 height: 170.72 } }
              text {
                rtf_data: "{\\rtf0\\ansi\\ansicpg1252...\\fs74\\...\\nosupersub Light guides my journey}"
              }
            }
            info: 3
          }
        }
      }
    }
  }
}

```

Important Notes

Adding an Additional Lyric Cue

If the new translation has more lyric sections than the original presentation, duplicate an existing complete `cues { ... }` block that has the desired bilingual layout.

1. Copy the entire cue, including its actions, slide, presentation, base slide, and both text elements.
2. Give the copied cue a new unique UUID. Do not reuse the original cue UUID.
3. Update the cue name or ordering metadata only when required by the presentation.
4. Change only the Indonesian and English lyric text after `\\nosupersub` in the copied cue.
5. Preserve every other value exactly, especially RTF formatting, font sizes, colors, bounds, positions, dimensions, and element metadata.

Do not build a new cue from scratch when a matching cue can be duplicated. This keeps the new section consistent with the existing presentation style.

DO's ✓

- ✓ Keep all RTF formatting codes intact
- ✓ Maintain exact spacing and indentation
- ✓ Keep font size codes (`\\fs104` , `\\fs74`) unchanged
- ✓ Keep Y position values for text alignment

- ✓ Always close with `}`
- ✓ Duplicate a complete cue when an additional lyric section is needed
- ✓ Assign a unique UUID to every duplicated cue
- ✓ Modify only the lyric text and necessary cue identity fields
- ✓ Test by encoding back to `.pro` format

DON'Ts ✗

- ✗ Don't remove RTF formatting codes
- ✗ Don't change font sizes, font numbers, or other styles
- ✗ Don't alter positioning, dimensions, or bounds
- ✗ Don't change color tables
- ✗ Don't remove escape characters (`\\`)
- ✗ Don't duplicate a cue with the same UUID

Encoding/Decoding Workflow

Decode Binary to Text (Read for editing)

```
protoc --decode rv.data.Presentation ./propresenter.proto < presentation.pro > decoded.txt
```

Make Edits

Edit the `decoded.txt` file with your bilingual lyrics

Encode Text Back to Binary (Save changes)

```
protoc --encode rv.data.Presentation ./propresenter.proto < decoded.txt > presentation.pro
```

Common Issues & Solutions

Issue 1: Text Not Appearing

Cause: Missing `\nosupersub` marker **Fix:** Ensure format is `\\nosupersub TextHere}`

Issue 2: Font Too Large/Small

Cause: A style or formatting value was changed while editing the lyric text **Fix:** Restore the original font size and all other style values. The objective is to change the lyrics, not the presentation design.

Issue 3: Text Overlap

Cause: Y coordinates not properly set **Check:**

- First element: `y: 869.1`
- Second element: `y: 920.3`
- Don't decrease the gap below 50 points

Issue 4: Encoding Fails

Cause: Malformed RTF or missing brackets **Fix:**

- Verify closing `}` is present
- Check backslashes are escaped (`\\` not `\`)
- Validate parentheses matching in RTF

Quick Reference: Text Replacement Pattern

```
FIND:      rtf_data: "{\\rtf0\\ansi\\ansicpg1252{...\\nosupersub [ORIGINAL_TEXT]}"}"
REPLACE:   rtf_data: "{\\rtf0\\ansi\\ansicpg1252{...\\nosupersub [NEW_TEXT]}"}"
```

File Location Reference

- **Input file:** `decoded_lyrics_<FILE_NAME>.txt`
- **Output file:** `<FILE_NAME> TRANSLATED.pro` (after encoding)
- **Proto definition:** `propresenter.proto`
- **Working directory:** `proto/autogen-proto/`

Validated Newer Context

Locate Secondary Placeholders

Use the `rtf_data` fields as the editing boundary. The primary block is identified by `fs104`; the secondary block is identified by `fs74`. Replace only text after `\\nosupersub` in the secondary block.

For a template containing literal placeholders, search specifically for the placeholder inside `rtf_data`:

```
grep -n '<TEXT>' <presentation-name>.txt
```

Do not replace a primary `fs104` lyric. Do not change RTF commands, font sizes, colors, bounds, positions, UUIDs, or element metadata.

Verified Translation Example

For example, when a cue's primary lyric is in Indonesian, translate it to English and place the result in its paired secondary RTF block:

```
Rise, proclaim the name of Jesus
```

The same rule applies independently to every cue: preserve the primary text and add only its matching secondary translation. When the primary lyric is English, translate it to Indonesian instead.

Encode And Verify

Encode the edited text to a new file, then decode that output again for verification:

```
protoc -I ./proto/autogen-proto --encode rv.data.Presentation ./proto/autogen-proto/propresenter.proto < ./<presentation-name>.txt > "./<presentation-name> TRANSLATED.pro"
protoc -I ./proto/autogen-proto --decode rv.data.Presentation ./proto/autogen-proto/propresenter.proto < "./<presentation-name> TRANSLATED.pro" > ./verified_<presentation-name>.txt
```

Recommended checks:

```
grep -c 'fs74.*nosupersub ' verified_<presentation-name>.txt
grep -c 'fs74.*nosupersub}' verified_<presentation-name>.txt
grep -c '<TEXT>' verified_<presentation-name>.txt
```

The first count should equal the number of lyric cues. The second and third counts must both be zero. Also compare the original and translated `fs104` RTF lines to confirm that primary lyrics were preserved.

Cleanup After Successful Encoding

After the encoded `.pro` file has been verified, keep the translated `.pro` file and remove temporary decoded text files when they are no longer needed:

```
rm ./<presentation-name>.txt ./verified_<presentation-name>.txt
```

On Windows Command Prompt, use:

```
del <presentation-name>.txt verified_<presentation-name>.txt
```

In the validation run, the translated output encoded to 24,463 bytes. `protoc` reported only the existing unused-import warning for `customOptions.proto`; encoding and re-decoding completed successfully.

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